EXPERIMENT 8

Aim: Configuring a Cryptocurrency Wallet using Metamask and integrating it with a DApp

THEORY

1. Cryptocurrency Wallet

A cryptocurrency wallet is a digital application used to manage blockchain accounts. It stores private keys that allow
users can access their cryptocurrency and authorize transactions.

2. MetaMask

MetaMask is a cryptocurrency wallet and browser extension that allows users to interact with Ethereum networks
and decentralized applications (DApps). It manages accounts, displays balances and signs transactions.

3. Ethereum Sepolia Test Network

Sepolia is an Ethereum test network used for testing blockchain applications and transactions without spending real
ETH. Test ETH can be obtained through faucets.

4. Ethereum Wallet Address

A wallet address is a unique identifier of an Ethereum account. It is used to send and receive ETH and interact with
smart contracts.

5. Decentralized Application (DApp)

A DApp is an application that interacts with blockchain technology. It uses a frontend interface, a Web3 provider
and a blockchain network to perform decentralized operations.

6. Wallet-DApp Integration

Wallet-DApp integration allows users to connect their MetaMask wallet to a web application. After permission is
granted, the DApp can access the wallet address and request blockchain transactions.

7. Ethers.js

Ethers.js is a JavaScript library used to connect web applications with Ethereum. It provides functions to access
wallet accounts, retrieve balances and send transactions.

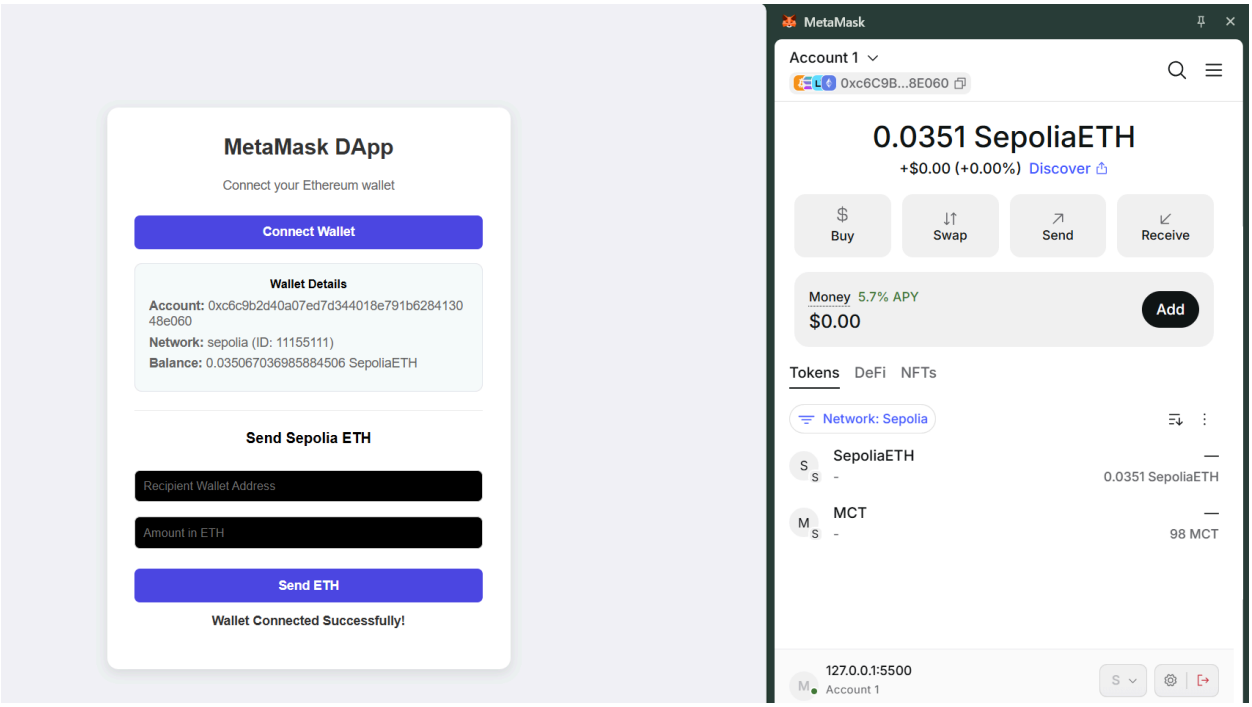
8. Gas Fees and Transactions

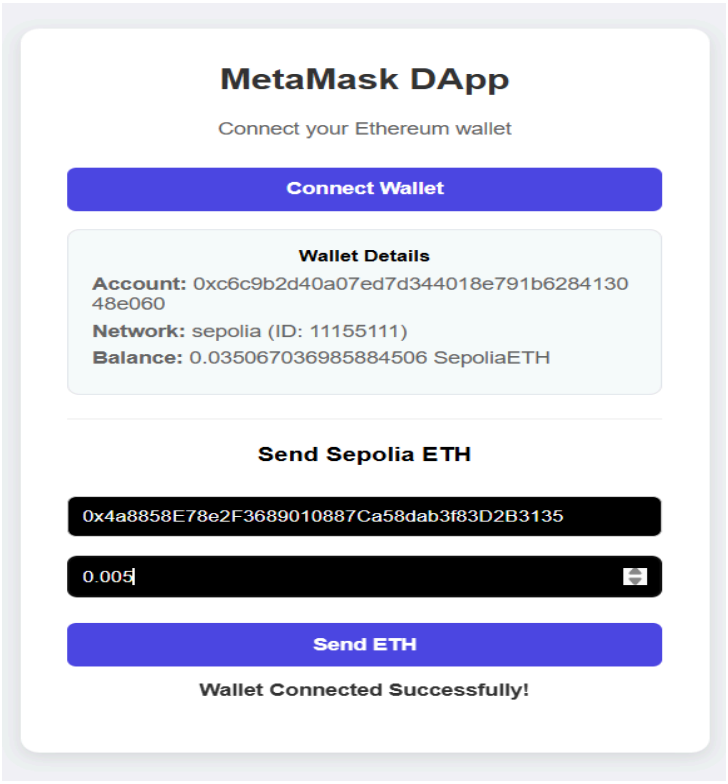
Gas fees are the transaction costs required to perform operations on the Ethereum network. MetaMask allows users
to review and approve transactions before they are submitted to the blockchain.

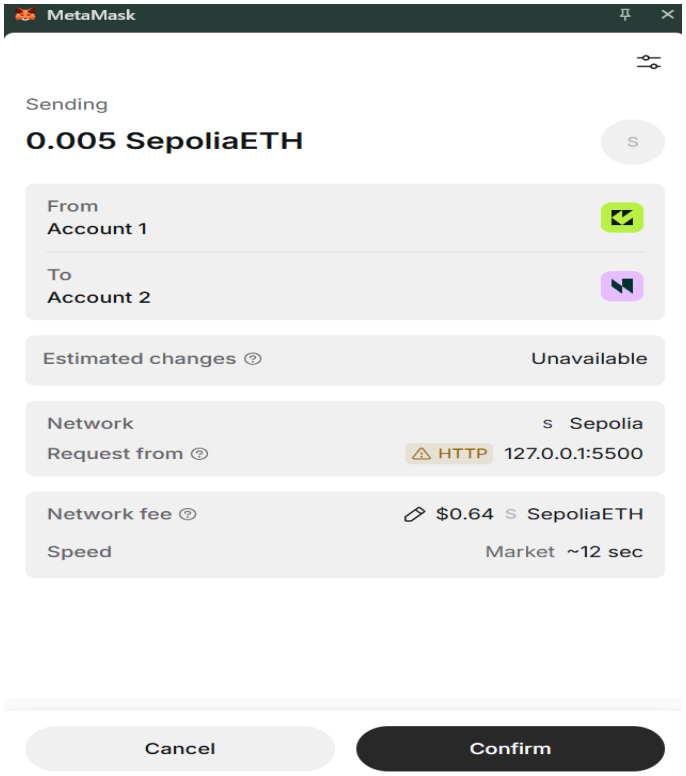
9. Blockchain Transaction Verification

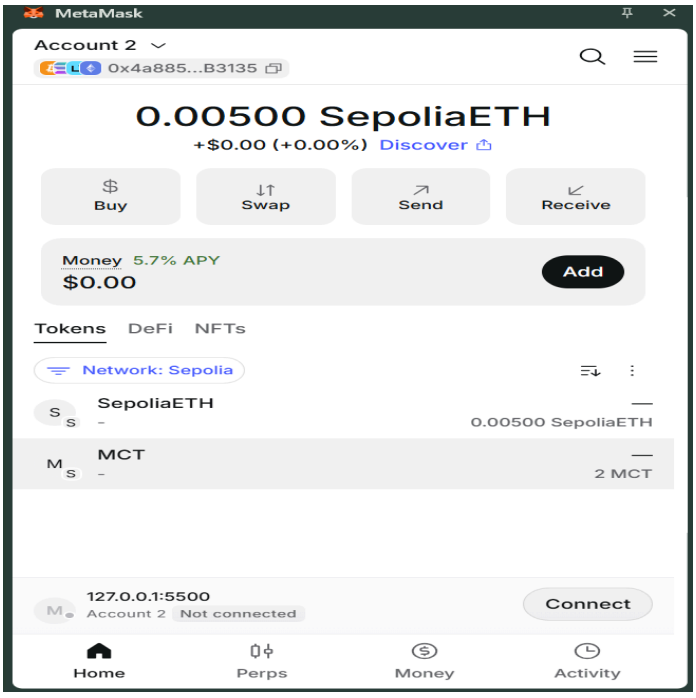
Transactions submitted to Ethereum are recorded on the blockchain. Sepolia Etherscan can be used to verify
transaction details such as sender, receiver, amount and transaction status.

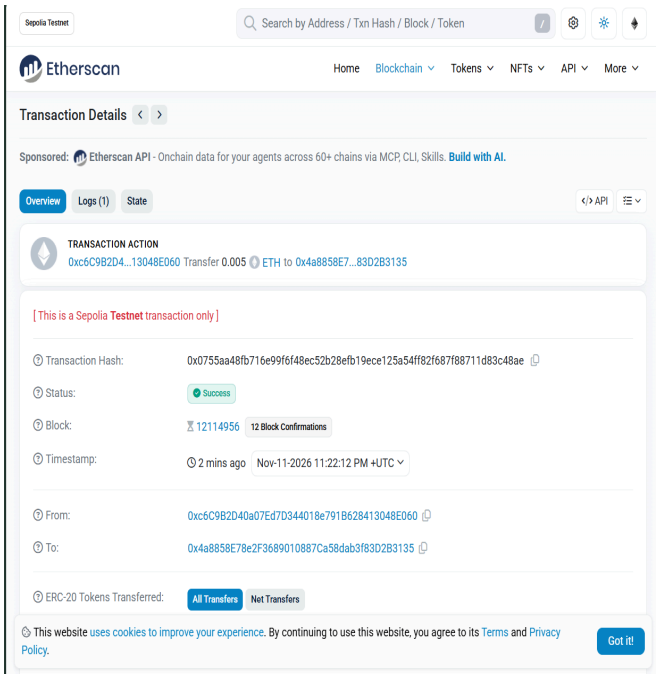
Output & Code:




MetaMask DApp
Connect your Ethereum wallet
Connect Wallet
Wallet Details
Account: 0xc6c9b2d40a07ed7d344018e791b628413048e060
Network: sepolia (ID: 11155111)
Balance: 0.035067036985884506 SepoliaETH
Send Sepolia ETH
0x4a8858E78e2F3689010887Ca58dab3f83D2B3135
0.005
Send ETH
Wallet Connected Successfully!


MetaMask
Sending
0.005 SepoliaETH
From
Account 1
To
Account 2
Estimated changes
Unavailable
Network
Sepolia
Request from
HTTP
127.0.0.1:5500
Network fee
$0.64 SepoliaETH
Speed
Market ~12 sec
Cancel
Confirm


MetaMask
Account 2
0x4a885...B3135
0.00500 SepoliaETH
+$0.00 (+0.00%) Discover
Buy
Swap
Send
Receive
Money 5.7% APY
$0.00
Add
Tokens
DeFi
NFTs
Network: Sepolia
SepoliaETH
0.00500 SepoliaETH
MCT
2 MCT
127.0.0.1:5500
Account 2 Not connected
Connect
Home
Perps
Money
Activity


Sepolia Testnet
Search by Address / Txn Hash / Block / Token
Etherscan
Home
Blockchain
Tokens
NFTs
API
More
Transaction Details
Sponsored: Etherscan API - Onchain data for your agents across 60+ chains via MCP, CLI, Skills. Build with AI.
Overview
Logs (1)
State
TRANSACTION ACTION
0xc6C9B2D4...13048E060 Transfer 0.005 ETH to 0x4a8858E7...83D2B3135
[This is a Sepolia Testnet transaction only]
Transaction Hash:
0x0755aa48fb716e99f6f48ec52b28efb19ece125a54ff82f687f88711d83c48ae
Status:
Success
Block:
12114956
12 Block Confirmations
Timestamp:
2 mins ago
Nov-11-2026 11:22:12 PM +UTC
From:
0xc6C9B2D40a07Ed7D344018e791B628413048E060
To:
0x4a8858E78e2F3689010887Ca58dab3f83D2B3135
ERC-20 Tokens Transferred:
All Transfers
Net Transfers
This website uses cookies to improve your experience. By continuing to use this website, you agree to its Terms and Privacy Policy.
Got it!

Conclusion

Thus, MetaMask was successfully configured and connected to the Ethereum Sepolia test network. A
simple DApp was created and integrated with MetaMask using Ethers.js. ETH was successfully transferred between two accounts and verified using Sepolia Etherscan. The experiment demonstrated
wallet-DApp integration and blockchain transaction processing